

Problem 1)

e) Can you further reduce model size beyond other parts? If yes, implement your proposed method and compare results. If not, explain why.

Yes we can combine the int8 quantization for the weight and activation while having knowledge distillation method to reduce our model size even more.

Problem 2)

- 1) What types of “Learning Blocks” are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self-supervised learning, etc.)?

Under the free tier edge impulse account for learning block, there are classification, regression, anomaly detection (GMM), anomaly Detection (K-means), anomaly detection.

- 2) List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

There are six classification layers:

1. Dense (Fully Connected Layer)

Basic neural-network layer where every neuron connects to every neuron in the next layer.

Purpose: learn relationships in processed DSP features (MFCC, spectral, statistical).

2. 1D Convolution / Pooling

Convolution + optional pooling along one dimension (time or sensor axis).

Purpose: learn local temporal/spatial patterns in raw signals or MFCC sequences.

3. 2D Convolution / Pooling

Convolution + pooling across two dimensions (time × frequency).

Purpose: learn spatial patterns in MFCC grids, spectrograms, or image-like data.

4. Reshape

Changes data shape like turning 1D to 2D

Purpose: prepare DSP output for convolution layers.

5. Flatten

Converts multi-dimensional tensors into a single vector.

Purpose: connect convolution layers to Dense layers.

6. Dropout

Randomly removes neurons during training.

Purpose: reduce overfitting and improve generalization.

- 3) What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse? List all available methods.

There are two types of model optimization methods, one of them is float 32 unoptimized and another quantized int 8 for a smaller model size where both are under EON compiler.

- 4) What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications.

There are two free tier versions of edge impulse with images and audio input where images mostly have the synthetic generation blocks of Dall-E images and Flux pro image generations where audio has whisper keyword spotting generation and eleven Labs synthetic audio block. Synthetic data is important in machine learning as it helps to fill out the dataset gaps and improve the model functionality and robustness with edge cases and less biases from training samples.